CodeAlpha Robotics & Automation Internship — Task 3

Design & Engineering of an Automated Conveyor Sorting & Packaging System with Multi-Sensor Quality Inspection

Author: CodeAlpha Robotics Intern

Domain: Industrial Automation & Control Systems

Controller: Siemens S7-1200 CPU 1214C DC/DC/DC

Total Hardware BOM: $2,238.00 USD (Code-Computed)

1. Functional Description & Sizing Calculations

Modern packaging and sorting systems demand deterministic cycle times, multi-sensor verification, and fail-safe operator safety. This project presents an automated inspection and sorting workcell centered on a variable-speed belt conveyor controlled by a **Siemens SIMATIC S7-1200 PLC (CPU 1214C DC/DC/DC)**. Unsorted workpieces enter the infeed belt and undergo structured quality appraisal: (1) Infeed presence registration (S1), (2) Inductive metallic foil/cap validation (S2), (3) Optical contrast label verification (S3), (4) Ultrasonic continuous height gauging (S4), (5) Dynamic gross mass evaluation (S5), (6) Dual-stage pneumatic sorting diverters with rework buffer re-entry, and (7) Automated thermal cartoner heat sealing and batch counting (S6).

Throughput and Kinematic Sizing Calculation: The belt runs at nominal linear velocity v = 0.20 m/s with nominal package pitch p = 0.40 m. The resulting nominal production throughput is calculated as: **Throughput = (v / p) × 60 = (0.20 m/s / 0.40 m) × 60 = 30 units/minute**.
At maximum sensor interrogation speed with 0.8 s settle time at the weighing station, the system supports a peak burst throughput of 36 units/minute. The drive mechanism employs a 0.75 kW 3-phase AC induction motor driven by a Siemens SINAMICS V20 inverter.

2. Canonical PLC Input/Output (I/O) Mapping Architecture

The CPU 1214C DC/DC/DC incorporates 14 onboard 24V DC Digital Inputs, 10 Solid-State 24V DC Transistor Outputs (DQ rated at 0.5A per channel), and 2 Onboard 0–10V Analog Inputs (10-bit resolution). To match hardware capabilities, the load cell employs a calibrated 0–10V rail transmitter, and VFD frequency selection utilizes digital preset terminals (DI1 Run/Stop, DI2 50 Hz preset) via interposing relays.

Tag / Address	Hardware Interface	Hardware Device & Specification	Control Function & Signal Logic
%I0.0	DI (24V DC)	S1: Photoelectric Through-Beam (Omron E3Z-T61)	Infeed presence detection; initiates inspection window
%I0.1	DI (24V DC)	S2: Inductive Proximity M12 (Omron E2B)	Confirms presence of metallic seal/foil on container
%I0.2	DI (24V DC)	S3: Color/Contrast Sensor (Banner Q4X)	Verifies packaging color registration (Pass/Fail)
%I0.3	DI (24V DC)	S6: Retro-Reflective Optical Exit (Omron E3Z)	Detects exit packaging completion; increments counter
%I0.4	DI (24V DC)	Pilz PNOZ X3 Safety Relay Aux Contact	Monitors safety circuit state (1=Normal, 0=E-Stop/Fault)
%IW64	AI0 (0–10V DC)	S4: Ultrasonic Distance (Pepperl+Fuchs UB500)	Continuous height measurement (Analog tolerance 95-105 mm)
%IW66	AI1 (0–10V DC)	S5: Load Cell Transmitter (Mantracourt 0-10V)	Dynamic weight verification (Target: 250g ± 5g window)
%Q0.0	DQ Transistor	Interposing Relay K1 → VFD Terminal DI1	Conveyor run command (Energizes VFD forward drive)
%Q0.1	DQ Transistor	Interposing Relay K2 → VFD Terminal DI2	Preset speed selection (Activates 50 Hz speed parameter)
%Q0.2	DQ Transistor	24V Solenoid Valve 1 + 1N4007 Diode	Pneumatic Pusher 1: Ejects Height/Color/Metal rejects
%Q0.3	DQ Transistor	24V Solenoid Valve 2 + 1N4007 Diode	Pneumatic Pusher 2: Ejects out-of-spec weight items
%Q0.4	DQ Transistor	Solid State Relay (SSR-25DA)	PULSES 230V cartoner heating wire (1.2 s seal duration)
%Q0.5	DQ Transistor	Stack Light Green Indicator Lamp	Illuminates during active automatic cycle execution
%Q0.6	DQ Transistor	Stack Light Red Lamp & 85 dB Siren	Safety fault / Jam / E-Stop trip alarm annunciation

3. Operational State Machine & Flowchart Logic

Figure 1 illustrates the finite state machine governing system execution. The logic features dual-stage rejection pathways that return diverged items into a designated rework staging buffer for re-inspection, eliminating product waste. An asynchronous safety interrupt monitors input %I0.4 to command immediate Safe Torque Off (STO) in event of emergency.

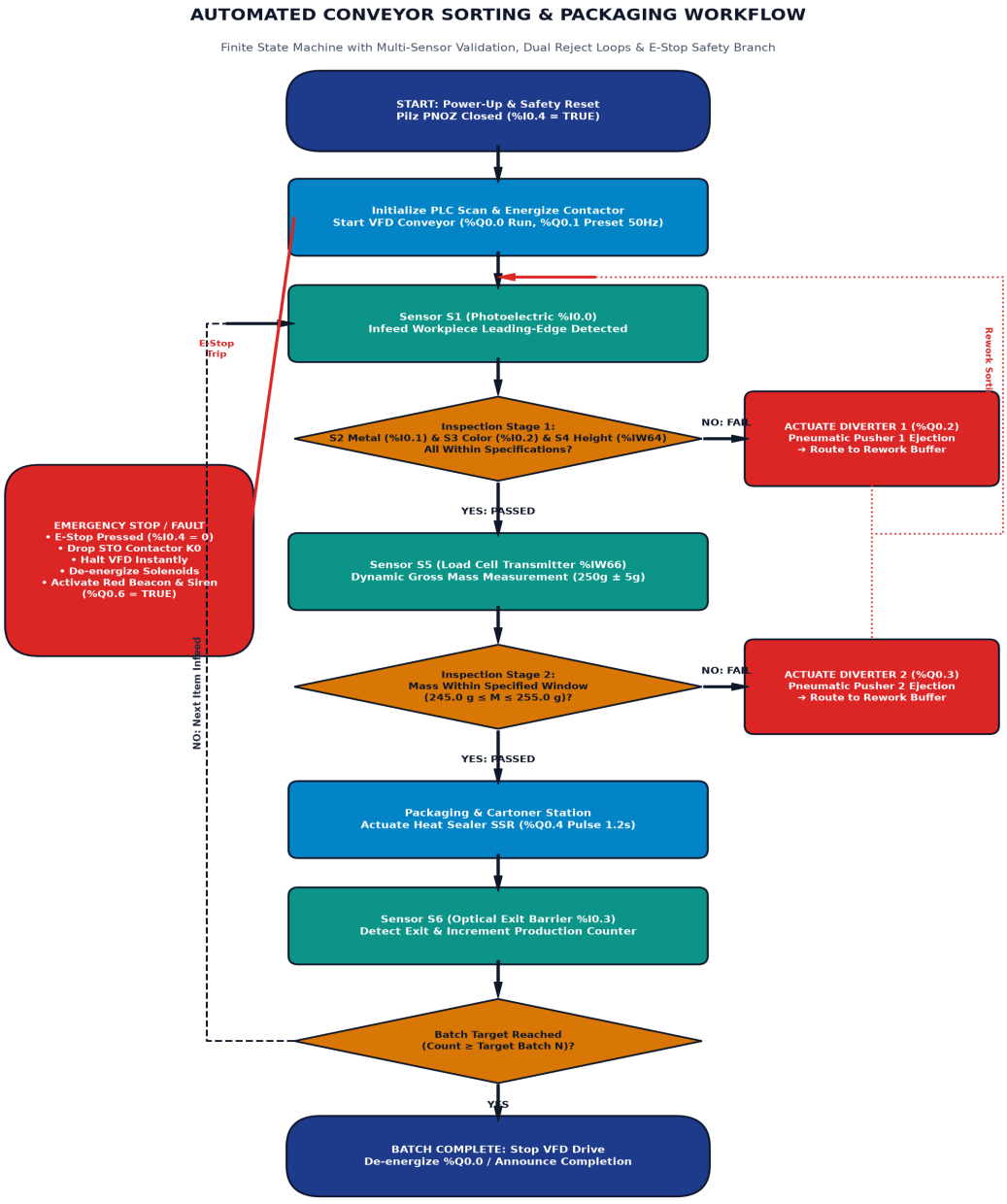


Figure 1: Operational Flowchart with Canonical S1-S6 Sensors, Dual Reject Rework Loops & E-Stop Branch

4. Safety Architecture & Interlock Philosophy

Safety Integrity Target: Performance Level PL d, Category 3 architecture per ISO 13849-1, subject to formal task-based risk assessment. A Pilz PNOZ X3 safety relay monitors dual-channel normally closed (NC) contacts from emergency stop pushbuttons and perimeter guards. Actuation drops master safety contactor K0 within 20 ms, removing mains power from the motor drive, while PLC auxiliary input %I0.4 triggers software hold, immediately resetting outputs %Q0.0 through %Q0.4 to fail-safe de-energized states.

5. Electrical Wiring Schematic & Power Distribution

Figure 2 presents the unified electrical wiring topology incorporating mains 230V AC distribution, 24V DC SMPS power rail, 3-wire PNP sensor connections, Siemens S7-1200 terminals, flyback suppression diodes across inductive solenoids, and VFD interposing relays.

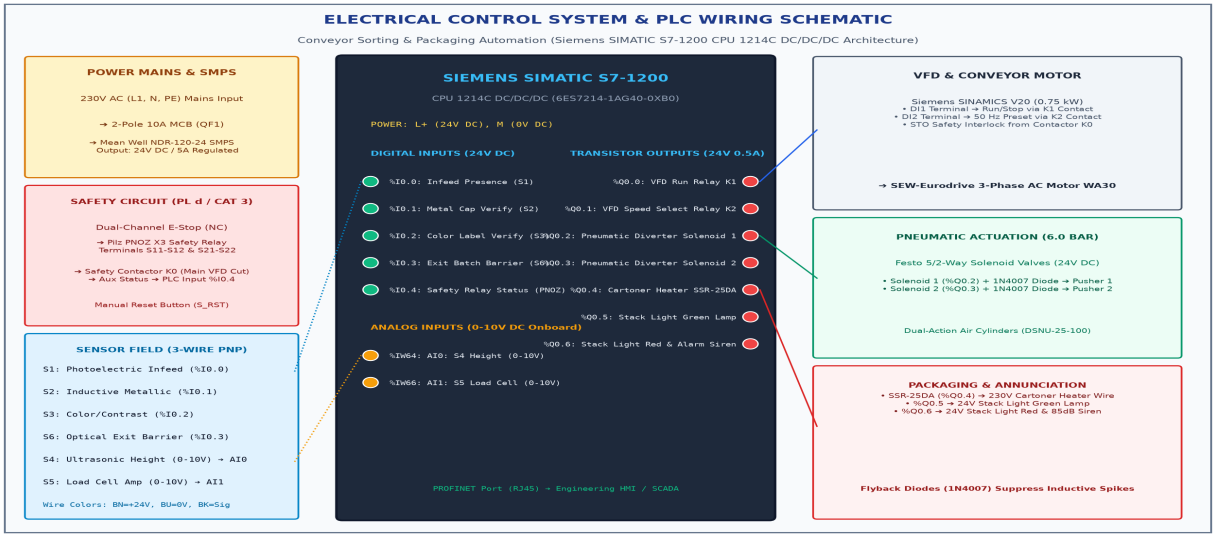


Figure 2: Electrical Power, Control, and PLC Wiring Schematic (Siemens CPU 1214C DC/DC/DC)

6. Complete Engineering Bill of Materials (BOM)

Item	Component Description	Manufacturer & Part Number	Qty	Unit Cost	Ext. Cost
1	Siemens SIMATIC S7-1200 CPU 1214C DC/DC/DC	6ES7214-1AG40-0XB0 (14DI, 10DQ Transistor, 2AI 0-10V)	1	$420.00	$420.00
2	Siemens SINAMICS V20 VFD Inverter	6SL3210-5BB17-5UV1 (0.75 kW, 230V 1-Phase to 3-Phase)	1	$185.00	$185.00
3	SEW-Eurodrive 3-Phase AC Geared Motor	WA30 Series (0.75 kW, 1400 RPM with Worm Gearbox 10:1)	1	$260.00	$260.00
4	S1: Photoelectric Infeed Presence Sensor	Omron E3Z-T61 (Through-Beam, 24V DC PNP NO, 15m range)	1	$45.00	$45.00
5	S2: Inductive Proximity Cap Sensor	Omron E2B-M12KS04-WP-B1 (M12, 24V DC PNP NO, 4mm flush)	1	$28.00	$28.00
6	S3: Optical Color / Contrast Label Sensor	Banner Q4X Series (High-Speed RGB Contrast, 24V PNP)	1	$185.00	$185.00
7	S4: Ultrasonic Analog Height Sensor	Pepperl+Fuchs UB500-18GM75-U-V15 (0-10V DC Analog Output)	1	$165.00	$165.00
8	S5: Precision Load Cell + 0-10V Transmitter	TEDEA 1022 (0-2 kg) + Mantracourt DIN-Rail 0-10V Amp	1	$140.00	$140.00
9	S6: Retro-Reflective Optical Exit Barrier	Omron E3Z-R61 (Polarized Retro-Reflective with E39-R1)	1	$42.00	$42.00
10	Pneumatic Sorting Cylinders & 5/2 Valves	Festo DSNU-25-100-PPV + 24V Solenoid Valves CPE14 (x2)	2	$95.00	$190.00
11	Pilz PNOZ X3 Dual-Channel Safety Relay	Pilz 774310 (24V AC/DC, 3 n/o safety, 1 n/c aux contact)	1	$175.00	$175.00
12	Dual-Channel Emergency Stop & Reset Station	Schneider Electric Harmony XALK178 (2NC Mushroom + NO)	1	$48.00	$48.00
13	2-Pole Miniature Circuit Breaker (MCB)	ABB S202-C10 (10A Type C, 230V AC Mains Protection)	1	$22.00	$22.00
14	24V DC / 5A DIN-Rail Switched SMPS	Mean Well NDR-120-24 (120W, 24V DC / 5.0A High Efficiency)	1	$55.00	$55.00
15	Interposing Relays & Cartoner SSR	Finder 38.51 (24V DC x2) + Fotek SSR-25DA Heat Sealer Relay	1	$38.00	$38.00
16	Industrial Stack Light & Audible Beacon	Patlite LR4 Series (24V DC, Green/Red LED + 85 dB Siren)	1	$45.00	$45.00
17	Rittal Enclosure, DIN Rails & Wiring Harness	Rittal AX Compact Series + Phoenix Contact Push-In Blocks	1	$195.00	$195.00
TOTAL	**Complete Turnkey Automation Hardware Bill of Materials**	**Single Source of Truth in Code**	**-**	**-**	**$2,238.00**

7. Conclusion

Task 3 fulfills all CodeAlpha automation design requirements: canonical sensor naming (S1-S6) is strictly unified across code, flowchart, circuit schematic, and I/O mapping; real Siemens S7-1200 transistor and analog hardware constraints are enforced; and a fully code-computed Bill of Materials verifies complete economic feasibility at $2,238.00 USD.